Department of Computer Science & Engineering
Microprocessor & Computer Architecture
UE24CS251B

MPCA Mini-project

Title: Golden Dome - ESP32-CAM laser targeting system

Team Members:

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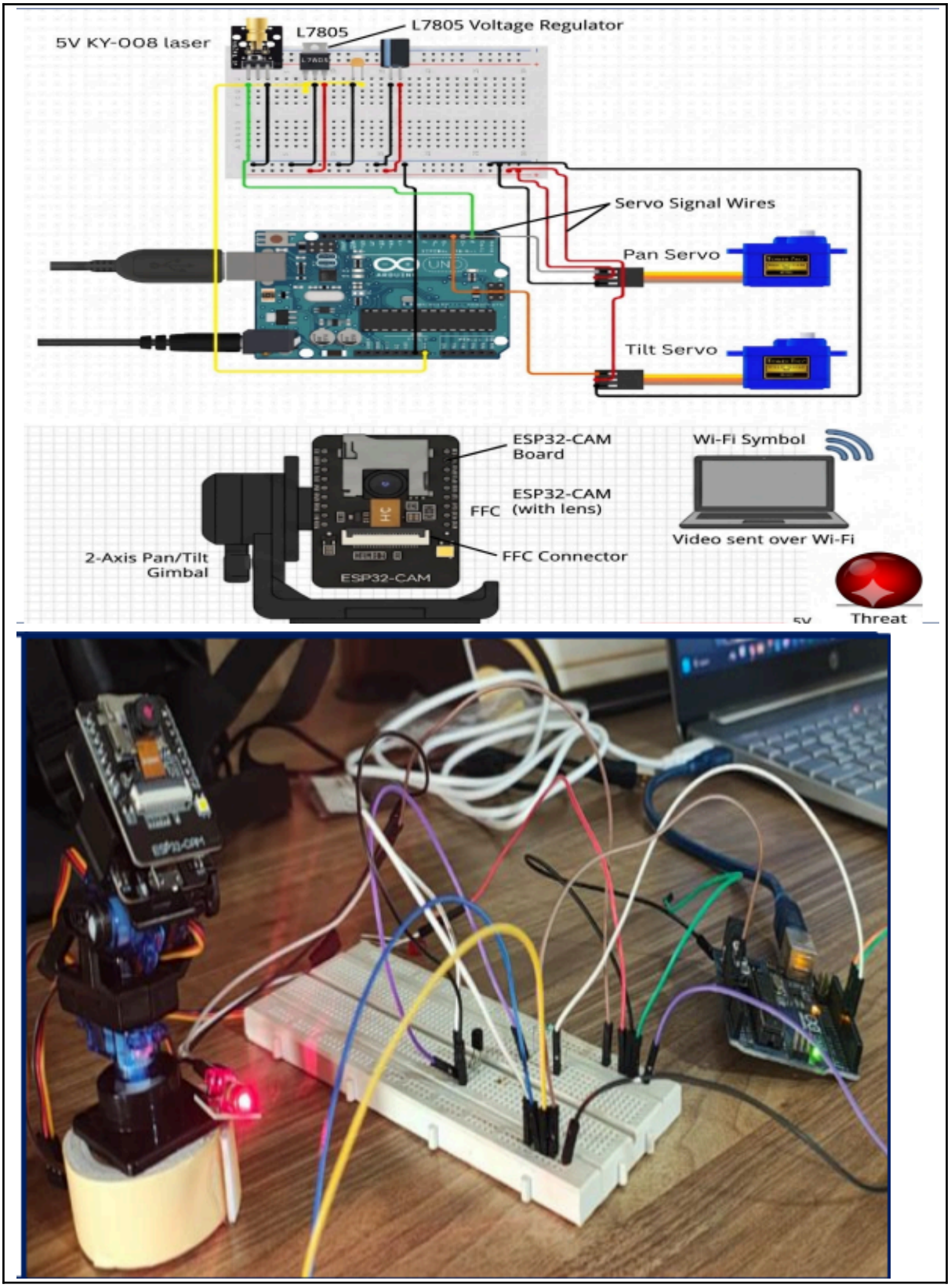
Abstract/Brief description of the project:

This project presents an **ESP32-CAM** based defense prototype for real-time **threat detection** and **tracking**. A live video stream is sent to a **FastAPI** server, where a red ball(missile prototype) is detected using computer vision.The target coordinates are calculated and angle of interception is sent to **servo motors** (2-axis servo bracket) , which aims a 5V **laser** towards the object.The system demonstrates a low-cost smart surveillance and defense model using IoT backed by AI (**Computer Vision**) .

Components used:
Arduino Uno , ESP32-Cam , ESP32-MB , 5V KY008 laser , 1k ohm resistor , 5V External power supply, 2N2222 NPN transistor , 2x SG90 servo motors , Two-Axis Pan and Tilt Servo Bracket , Jumper Wires , FastAPI backend on laptop

Circuit Diagram and Prototype Picture:


5V KY-008 laser
L7805
L7805 Voltage Regulator
Servo Signal Wires
Pan Servo
Tilt Servo
ESP32-CAM Board
ESP32-CAM (with lens)
FFC
FFC Connector
2-Axis Pan/Tilt Gimbal
ESP32-CAM
Wi-Fi Symbol
Video sent over Wi-Fi
Threat

Thank you